Woohyun Song

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Interested in using machine learning to find predictable structure and signals in noisy, high-variance real-world data.

Education

Seoul National University — B.S. in Computer Science and Engineering *Mar 2022 – Present*

Projects & Research

1st STATIZ KBO Outcome Prediction Competition *Mar 2026 – Present*

- Built an **end-to-end ML system** predicting home-team win probability for KBO regular-season games and auto-submitting daily picks via the competition API, deployed on AWS EC2 (collection → feature build → training → prediction → submission).
- Designed **31 features** normalized as home–away differences, adapting a published KBO win-prediction methodology: starting-lineup OPS-sum gap, starters' blended-FIP gap, key-bullpen fatigue gap, and more.
- Built a **RandomForest + ElasticNet Logistic Regression ensemble** trained on ~2,410 games (2023–2025 seasons) and validated with expanding-window backtesting; self-built an HTML observability dashboard for per-game prediction review.
- Ranked **3rd overall out of ~50 teams** (as of May 28, 2026).

International Quant Championship (IQC) 2026 — WorldQuant BRAIN *Mar 2026 – Present*

- Developed systematic alpha signals as a team in a global alpha-research competition with **100,000+ teams**.
- Engineered reversion and news-momentum alphas from alternative data (social-media sentiment, news-flow signals).
- Used LLMs to accelerate hypothesis generation and alpha-expression iteration.
- **Ranked global #95 / Korea #7** on the in-sample leaderboard; **advanced to Stage 2**.

Analyzing Effective Intentional Walk Strategies in KBO Using ML *Mar 2023 – Jun 2023*

- First independent project, undertaken with no prior background in AI; **manually collected and processed raw data** due to the lack of existing datasets, gaining hands-on experience in feature definition.
- Applied logistic regression, decision tree, and SVM to analyze strategic variables (pitcher ERA, base-out state, OPS).
- Top project for **Scientific Writing** (L0440.000900), graded **A+**.

Military Service

Republic of Korea Air Force (Tactical Communications Operator) *Feb 2024 – Nov 2025*

- As a member of the Soldier Autonomy Committee, planned and ran a peer Talent-Sharing Mentoring Program to expand self-development opportunities on base.
- Taught introductory algorithms — graph theory, dynamic programming, backtracking, binary search — distributing practice problems and fielding questions from fellow soldiers.

Relevant Coursework

Introduction to Machine Learning (Spring 2026, in progress) · Computational Linguistics (Spring 2026, in progress) · Artificial Intelligence (Fall 2023) · Linear Algebra (Winter 2025)

Skills

Programming Languages: Python, C++
ML & Data: pandas, NumPy, scikit-learn
Certifications: SQLD (SQL Developer), TOEIC 955